

PROMPT ENGINEERING TECHNIQUES

1. Weak Prompt

You are a software engineer.
Review this code for bugs:

```
Python
def calculate(price, tax_rate):
    total = price + (price * tax_rate)
    return total
print(calculate(100, "0.18"))
```

There is a bug in the call to `calculate` :

```
</> Python
print(calculate(100, "0.18"))
```

`tax_rate` is passed as a string (`"0.18"`), but the function expects a numeric value.

What happens:

```
</> Python
price * tax_rate
```

becomes:

```
</> Python
100 * "0.18"
```

which produces the string `"0.18"` repeated 100 times. Then Python tries to execute:

```
</> Python
100 + "0.180.180.18..."
```

and raises:

```
</> Python
TypeError: unsupported operand type(s) for +: 'int' and 'str'
```

Fix

Pass `tax_rate` as a number:

```
</> Python
def calculate(price, tax_rate):
    total = price + (price * tax_rate)
    return total

print(calculate(100, 0.18))
```

Output:

```
</> Python
118.0
```

2. Strong Prompt: Priorities

You are a Director of Global Supply Chain Procurement. You know that accepting sudden price hikes destroys product margins, but walking away could halt production entirely.

Your priorities are:

Maintaining uninterrupted production continuity at all costs.

Protecting the company's Q3 profit margins.

Evaluating long-term contract leverage and alternative sourcing strategies.

Propose a step-by-step negotiation and operational action plan to handle a sudden 15% price hike from our primary raw material supplier.

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A sudden 15% price increase from a primary supplier requires an immediate response that balances supply continuity with financial performance. The first priority is ensuring there is no disruption to production by confirming inventory levels, safety stock, and upcoming delivery commitments. Maintaining supply is critical while negotiations are underway.

Next, request a detailed breakdown of the supplier's cost drivers and validate whether the increase is supported by market conditions. Compare the proposed increase against industry benchmarks, commodity indices, and alternative supplier pricing to understand the true extent of the cost pressure.

During negotiations, avoid accepting the full increase immediately. Propose alternatives such as a phased increase, temporary surcharge, volume commitments, or a longer-term contract in exchange for improved pricing. The goal is to share risk while preserving the relationship and protecting margins.

At the same time, begin evaluating alternative sourcing options. Issue RFQs to qualified suppliers, assess their capabilities, and conduct quality and compliance reviews. Even if switching is not immediate, creating competition improves negotiating leverage and reduces future dependency on a single supplier.

To protect Q3 profitability, work with operations and finance teams to identify cost-reduction opportunities, improve production efficiency, and evaluate selective customer price adjustments where contracts permit. Focus available supply on higher-margin products if material availability becomes constrained.

For the long term, strengthen procurement resilience through dual-sourcing strategies, strategic inventory buffers, and contracts that include price-adjustment formulas, escalation caps, and regular review mechanisms. This reduces exposure to unexpected supplier price increases and improves negotiating power in future discussions.

Key Actions:

- Secure supply continuity and assess inventory coverage.
- Validate the supplier's cost increase with market data.
- Negotiate a phased or reduced price increase.
- Use contract leverage and volume commitments strategically.
- Qualify alternative suppliers to reduce dependency.
- Implement margin-protection and cost-saving initiatives.
- Establish long-term sourcing and contract risk controls.

3. Strong Prompt: Role

You are an expert Product Marketing Manager. You know that users don't care about a feature unless they understand the immediate benefit to their daily life.

Write a feature announcement email about our new "Dark Mode" setting. Focus heavily on user health, reducing late-night eye strain, and saving phone battery life, rather than just stating that the color changed.

What the AI will do: It will frame the update as a wellness and battery-saving upgrade, making it sound exciting and essential.

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Subject: New Dark Mode: Easier on Your Eyes, Better for Your Battery

Hi there,

You asked for a more comfortable way to use the app at night, and it's here.

Introducing **Dark Mode** — a new display option designed to help make late-night browsing easier on your eyes while also helping conserve battery life on supported devices.

Why you'll love it

👁️ Reduce late-night eye strain

Bright screens can feel harsh in dark environments. Dark Mode provides a softer viewing experience that can feel more comfortable when you're checking in during the evening or early morning.

🔋 Help your battery last longer

On many modern phones, darker interfaces can use less power, helping you get more from your battery throughout the day.

🌙 A more comfortable everyday experience

Whether you're catching up before bed, commuting in low light, or simply prefer a darker look, Dark Mode helps create a more relaxed viewing experience.

Turn it on in seconds

Go to **Settings** → **Appearance** → **Dark Mode** and switch it on whenever you like.

We're always working to make the app more comfortable and enjoyable to use, and we hope Dark Mode becomes one of your favorite updates yet.

Thanks for being with us,

The Product Team

4. Strong Prompt: Role + Priorities

You are a senior financial analyst with 10 years of experience. You focus on identifying risks, anomalies, and trends that the leadership needs to act on - not just reporting numbers.

Analyze this quarterly revenue and marketing spend data. Do not just summarize the numbers; pinpoint the critical efficiency anomaly and explain the hidden risk to our current growth strategy:

Q1: \$120,000 (Marketing: \$20,000)

Q2: \$145,000 (Marketing: \$22,000)

Q3: \$110,000 (Marketing: \$45,000)

Q4: \$160,000 (Marketing: \$25,000)

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The critical anomaly is Q3:

- Q1 efficiency: **\$6.00 revenue per \$1 marketing**
- Q2 efficiency: **\$6.59 per \$1**
- Q3 efficiency: **\$2.44 per \$1** ← major deterioration
- Q4 efficiency: **\$6.40 per \$1**

Despite **doubling marketing spend** from \$22k to \$45k in Q3, revenue **fell** from \$145k to \$110k. That suggests a severe drop in marketing effectiveness, targeting quality, conversion rates, or customer demand.

Hidden risk: Leadership may focus on the strong Q4 revenue recovery (\$160k) and overlook the fact that growth appears fragile. If revenue can collapse while marketing investment rises, the business may not have a predictable acquisition engine. Without understanding what caused Q3, future growth forecasts and marketing budgets could be based on misleading assumptions about scalability and ROI.

Priority action: Investigate Q3 channel performance, conversion metrics, customer quality, and market conditions before increasing marketing spend further.

5. Zero Shot + Structured Prompting

Premium Car Transport (High-Value Single Carrier)

Subject: Enclosed transport or a flatbed? Let's be real.

Body:

If you're moving a standard commuter hatchback across the state, call any generic towing company. A basic open flatbed will do the job, and it'll be cheap.

But if you're shipping a high-end luxury vehicle, a pristine classic, or a low-clearance sports car, open-air shipping is a gamble you shouldn't take. Highway debris doesn't care about your ceramic coating.

Our premium enclosed car carriers treat vehicles like cargo, not just a car tied to a truck. Fully covered walls shield the vehicle from road grime and weather. Hydraulic lift gates keep the loading angle perfectly flat so low bumpers never scrape. Plus, our transit insurance covers the actual value of the machine—not just a generic weight-based baseline.

It costs more than an open trailer. We aren't going to pretend it doesn't. But fixing a rock chip on custom factory paint costs a whole lot more.

[Get an Enclosed Transport Quote →]

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This copy is effective because it sells risk reduction rather than transportation.

What works well:

- Opens with a strong qualifier, immediately separating budget buyers from premium customers.
- Highlights specific risks (road debris, weather, scraping) that luxury and classic car owners care about.
- Uses concrete benefits: enclosed carrier, hydraulic lift gates, and higher-value insurance coverage.
- Acknowledges the higher price instead of avoiding it, which builds credibility.
- Ends with a cost-of-damage comparison that reinforces the value proposition.

Potential improvement:

The headline could be more benefit-focused. For example:

"Protect Your Luxury Vehicle with Enclosed Transport"

You could also add a trust signal such as:

- Number of vehicles transported
- Years in business
- Customer rating
- Specialized experience with luxury, exotic, or classic cars

Overall, the message positions enclosed transport as a protection and peace-of-mind investment, not just a shipping service.

6. Task Decomposition

Prompt 3 — Root cause diagnosis

Based on the connections you just identified, what is the root cause of this campaign's underperformance? I want your actual diagnosis, not a list of everything that went wrong.

Structure your thinking as:

PRIMARY CAUSE: The single biggest factor (with evidence from the data)

CONTRIBUTING FACTORS: 2-3 secondary issues that amplified the primary cause

WHAT WASN'T THE PROBLEM: 1-2 things that look bad on the surface but aren't actually root causes when you examine the data carefully

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Example Root Cause Diagnosis (using a hypothetical campaign dataset)

PRIMARY CAUSE:

Poor audience targeting was the primary driver of underperformance.

The data showed that 70% of spend was allocated to broad audiences that generated high impressions and clicks but very low conversion rates. While CTR remained healthy at 2.8%, the conversion rate was only 0.4%, indicating that the campaign attracted interest from users who were unlikely to buy. The gap between engagement and conversion is the strongest evidence that traffic quality—not ad visibility—was the core problem.

CONTRIBUTING FACTORS:

- **Weak landing page experience:** Bounce rates exceeded 75%, causing many interested users to leave before completing key actions.
- **Budget concentration in low-performing segments:** Spend continued flowing to audiences with high CPA while better-performing segments received limited budget.
- **Generic creative messaging:** Ads generated clicks but did not clearly communicate the product's value proposition, reducing conversion efficiency.

WHAT WASN'T THE PROBLEM:

- **Click-through rate (CTR):** Although overall results were poor, CTR was above industry benchmarks, showing that ads were attracting attention.
- **Budget size:** The campaign had sufficient spend to generate meaningful results. The issue was how the budget was allocated, not the amount invested.

Conclusion:

The data supports **poor audience targeting** more strongly than alternative explanations such as creative quality or budget limitations. Strong CTR combined with weak conversion performance indicates the campaign successfully attracted attention but from the wrong users. Fixing targeting would likely have produced the largest performance improvement.